

Homework 3

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Problem 1

For a point x and a non-empty subset of a metric space (X, d) , define $d(x, A) = \inf \{d(x, a) \mid a \in A\}$

- Prove that $d(x, A) = 0$ iff $x \in \overline{A}$
- Show that if y is another point in X then $d(x, A) \leq d(x, y) + d(y, A)$
- Prove that $x \mapsto d(x, A)$ gives a continuous map from X to $\mathbb{R}$

Solution. 1.

($\Rightarrow$)

Suppose we have x such that $d(x, A) = \inf \{d(x, a) \mid a \in A\} = 0$

Since the inf is zero we have that for all $\epsilon > 0$ there exists $a \in A$ such that $d(x, a) < \epsilon$

Then $x, a \in B_\epsilon(x)$ meaning $B_\epsilon(x) \cap A \neq \emptyset$, then from the ball characterization of the closure of a set $a \in \overline{A}$

($\Leftarrow$)

This case is just the reverse of the above

Suppose we have $x \in \overline{A}$ then for all $\epsilon > 0$ $B_\epsilon(x) \cap A \neq \emptyset$, i.e. for all $\epsilon > 0$ $d(x, a) < \epsilon$, so the inf of $\{d(x, a) \mid a \in A\}$ is zero, and hence $d(x, A) = 0$

□

Solution. 2.

Let $a \in A$ and $x, y \in X$

Then $d(x, A) \leq d(x, a)$ and by the triangle inequality $d(x, a) \leq d(x, y) + d(y, a)$

Then $d(x, A) \leq d(x, y) + d(y, a)$, rearranging we get $d(x, A) - d(x, y) \leq d(y, a)$.

Hence $d(x, A) - d(x, y)$ is a lower bound on $d(y, a)$, then $d(x, A) - d(x, y) \leq d(y, A)$, since $d(y, A)$ is the greatest lower bound.

Finally, we get $d(x, A) \leq d(x, y) + d(y, A)$

□

Solution. 3.

Let $f: X \rightarrow \mathbb{R}: f(x) = d(x, A)$

f is continuous iff for every $\epsilon > 0$ there exists a $\delta > 0$ such that $d(x, y) < \delta \Rightarrow |d(x, A) - d(y, A)| < \epsilon$

Choose $\delta = \epsilon$

Then from the last part we get

$$d(x, A) - d(y, A) < \delta = \epsilon, \quad d(y, A) - d(x, A) \leq d(x, y) < \delta = \epsilon$$

So $|d(x, A) - d(y, A)| < \epsilon$

Hence f is continuous

□

Problem 2

Prove that a finite subset of a metric space has no limit points

Solution. Let (X, d) be a metric space and $A \subset X$ a finite subset

Suppose there exists some limit point of A , say x

Then there exists some sequence of $\{x_n\}$ with $x_n \in A \setminus \{x\}$ and $x_n \rightarrow x$

Then for all $\epsilon > 0$ there exists some $N \in \mathbb{N}$ such that $d(x_n, x) < \epsilon$ for all $n \geq N$

Note: if $A \subset \{x\}$ then there doesn't exist any sequence as above and consequently there are no limit points

Choose $\epsilon_0 = \min \{d(a, x) \mid a \in A \setminus \{x\}\}$, since A is finite we have ϵ as the minimum of a finite set of distance with each not being zero, hence $\epsilon_0 > 0$

Then there is some N such that $d(x_n, x) < \epsilon_0$ for all $n \geq N$, but also $\epsilon_0 \leq d(x_n, x)$ since $x_n \in A \setminus \{x\}$

This is a contradiction, hence A has no limit points $\square$

Problem 3

Prove that a map $f: X \rightarrow Y$ of metric spaces is continuous iff for every subset $B \subset Y$ we have the $f^{-1}(\overset{\circ}{B})$ is contained in the interior of $f^{-1}(B)$

Solution. $(\Rightarrow)$ Assume $f: X \rightarrow Y$ is continuous, then we have that the preimage of all open sets are open

For all $B \subset Y$, we have that $\overset{\circ}{B}$ is open, hence $f^{-1}(\overset{\circ}{B})$ is open.

Also $\overset{\circ}{B} \subset B$, meaning $f^{-1}(\overset{\circ}{B}) \subset f^{-1}(B)$

So now, $f^{-1}(\overset{\circ}{B})$ is a subset of the preimage of B and also that it is open

So it is an open subset of $f^{-1}(B)$, from the definition of the interior, $\overset{\circ}{A} = \bigcup_{X \subset A \text{ open}} X$, we have that $f^{-1}(\overset{\circ}{B})$ is contained within the interior of $f^{-1}(B)$ $\square$

Problem 4

For a subset A of a metric space X , prove

- $\overset{\circ}{A} = A \setminus \partial A = \overline{A} \setminus \partial A$
- $\overline{X \setminus A} = X \setminus \overset{\circ}{A}$
- $\partial A = \overline{A} \cap \overline{X \setminus A} = \partial(X \setminus A)$
- ∂A is closed in X